

Do iron and oxygen stress trigger the same response in bacteria?

A side-by-side reanalysis of two published pathogens

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Background

Inside a human host, bacteria are starved of two things at once. Iron is withheld on purpose — a defence called nutritional immunity — while oxygen runs low in abscesses, mucus, biofilms and the gut lumen. Both shortages force a major metabolic rewiring, and both are tied to how virulent the organism becomes.

The two are chemically linked (Figure 1), and some bacteria wire that link into their regulation: in *Shigella*, the oxygen sensor ArcA controls Fur, the master iron regulator. Because of this, the literature often treats low iron and low oxygen as one shared host-stress program. Whether they actually move the same genes the same way has never been tested directly.

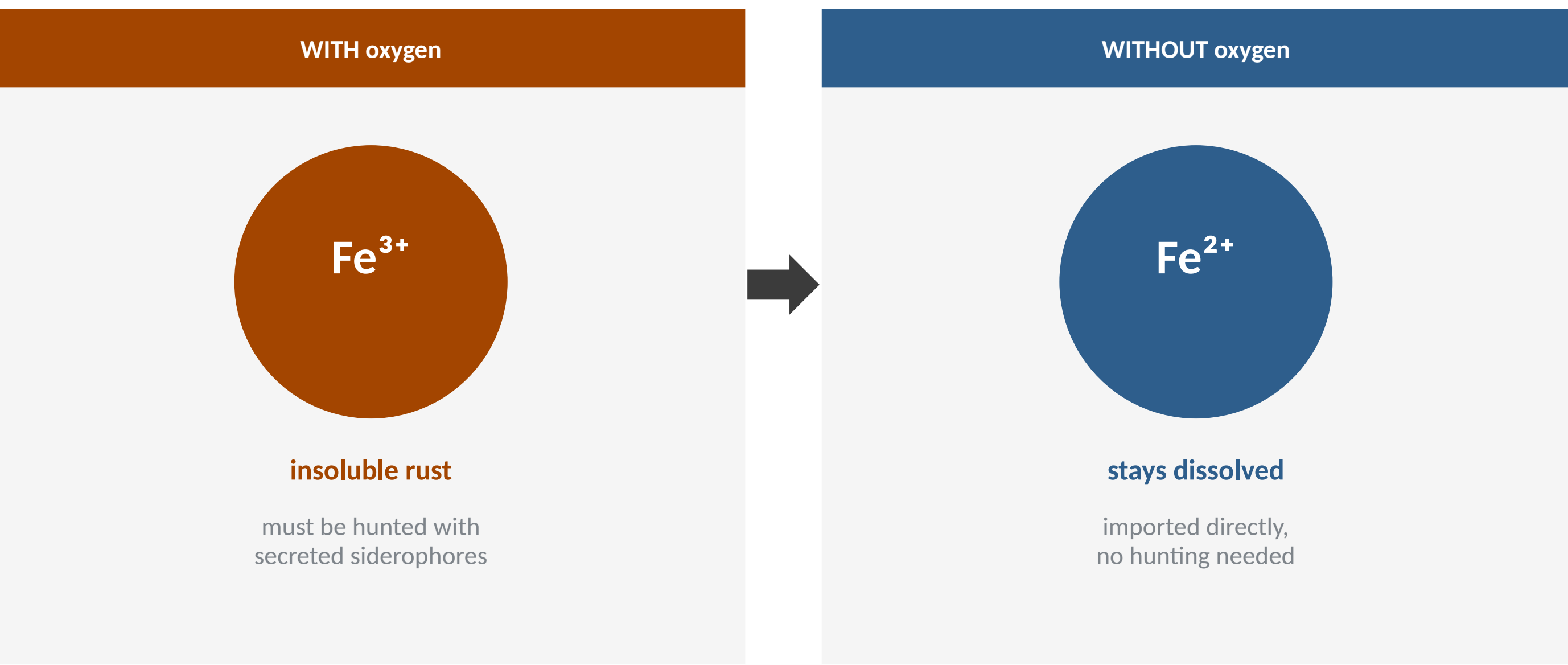


Figure 1. Removing oxygen makes iron easier to obtain, not harder. Any shared response between the two shortages should reflect this.

Methods

- No new experiments. Four published datasets were retrieved from GEO, ENA and journal supplements and re-analyzed under one common procedure.
- Two organisms from different bacterial families: *P. aeruginosa*, an environmental opportunist causing lung and wound infection, and *S. flexneri*, an obligate human gut pathogen.
- Microarrays were re-normalized from raw intensity files with RMA; RNA-seq arms use the original authors' published differential expression tables.
- Genes were sorted into seven functional categories. Definitions and cutoffs were written down and locked before any *Shigella* data were examined.
- For the *Shigella* iron arm the blinding was enforced in code: the assignment script loaded only identifier and annotation columns, never the fold-change column.
- Cutoffs: adjusted $p < 0.05$ and at least two-fold change. A direction is reported only when three or more genes support it.
- Gene sets were externally reviewed and revised three times, including a completeness audit that added 47 genes. No revision changed any direction call; all are published in a revision log.

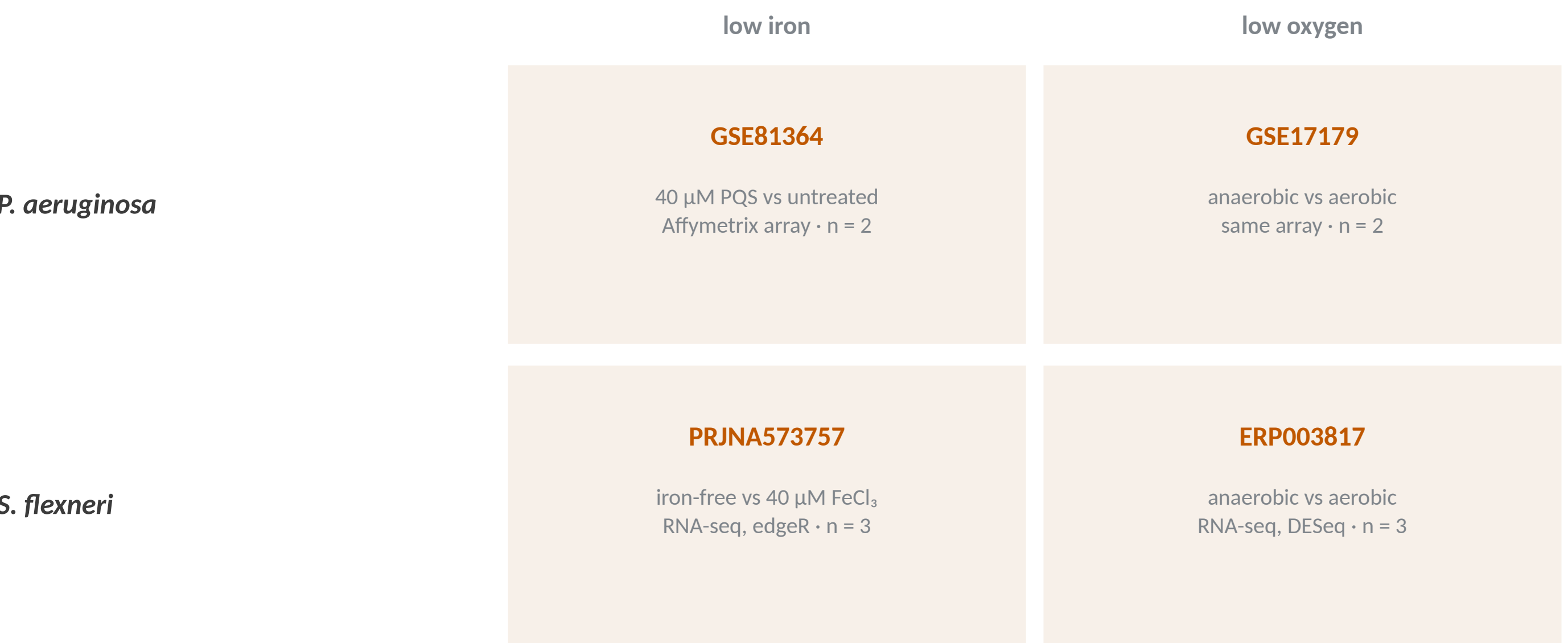


Figure 2. All four cells are filled, so organism and perturbation are never confounded. Most published comparisons fill only two.

Results

Both bacteria answer iron scarcity the same way: switch on the machinery that scavenges iron from outside the cell, switch off the protein that stores it inside. In *S. flexneri* 24 of 25 iron genes rose; in *P. aeruginosa*, 28 of 29. The organisms use different siderophores and different regulators, yet the logic is identical — acquire more, store less. The result survives dropping all fifteen uncharacterised iron transporters.

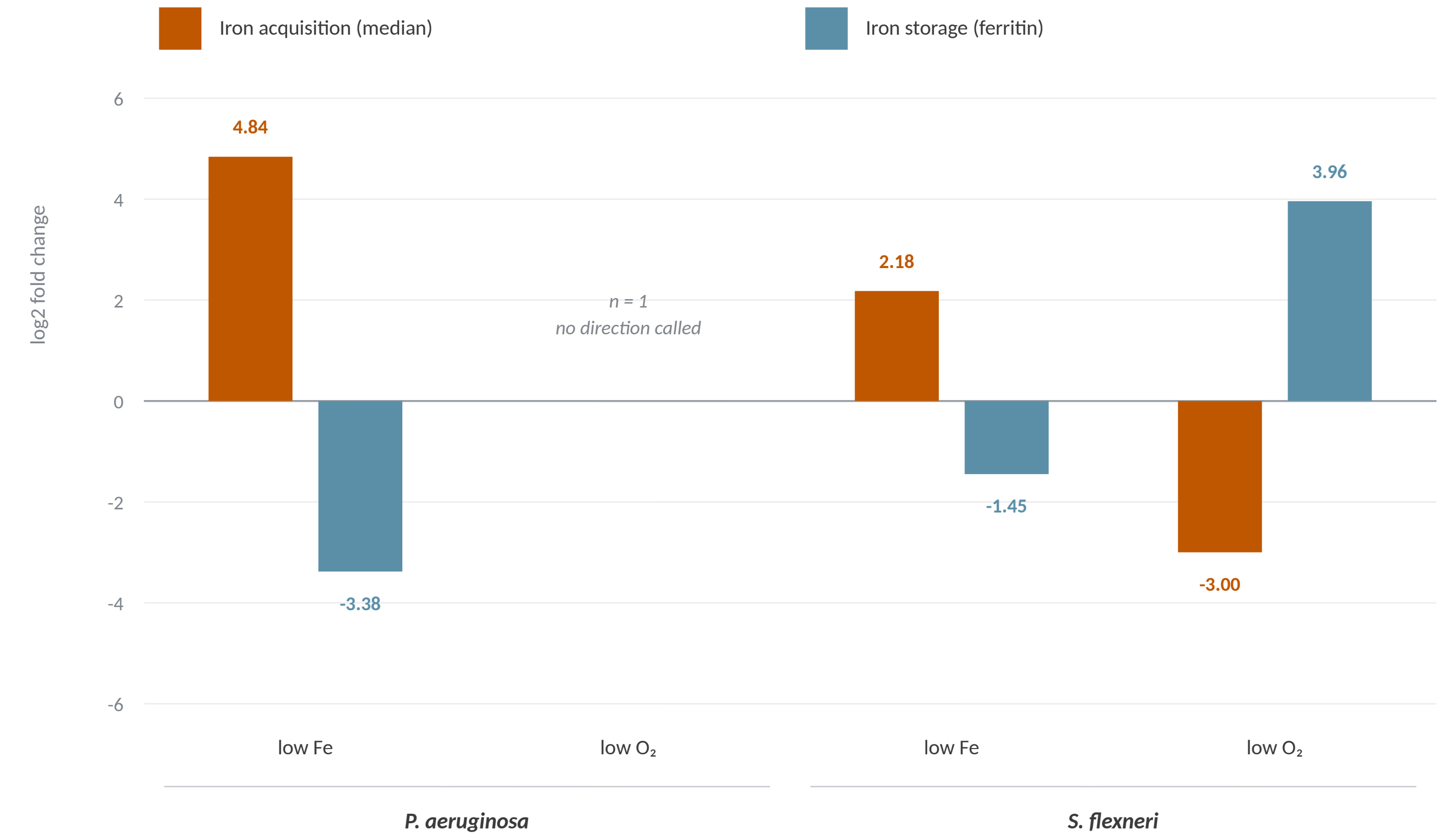


Figure 3. Iron limitation drives acquisition up and storage down in both species. Oxygen limitation reverses that in *S. flexneri* and does nothing in *P. aeruginosa* (one gene, no bar drawn).

Oxygen scarcity splits them. *S. flexneri* reverses its iron program — 20 of 22 genes fall and ferritin rises about 16-fold, consistent with ferrous iron becoming freely available. *P. aeruginosa* registers a single gene across the whole category, too few to call a direction at all.

	<i>P. aeruginosa</i>		<i>S. flexneri</i>	
	low Fe	low O ₂	low Fe	low O ₂
Iron uptake & storage	UP	too few	UP	DOWN
Anaerobic respiration	DOWN	UP	too few	UP
Oxidative stress	DOWN	mixed	UP	DOWN
Carbon / amino acid	DOWN	mixed	mixed	mixed
Secretion & virulence	UP	DOWN	UP	no data
Motility	too few	mixed	too few	too few
Quorum sensing / biofilm *	UP	too few	UP	no data

Figure 4. All seven categories. Row 2 is the internal control: anaerobic respiration rises without oxygen and falls without iron, as it must, since those enzymes are iron-dependent. That the categories recover known biology is what licenses reading the iron row above. * Not comparable — *S. flexneri* has no quorum-signal synthase.

Conclusion

- Iron shortage produces a conserved response. Two bacteria separated by hundreds of millions of years, using different siderophores and regulators, move the same category the same way.
- Oxygen shortage does not. Only the gut pathogen converts low oxygen into a major iron response.
- The two shortages are therefore not interchangeable, and describing them as one host-stress program blurs a real difference between organisms.
- The gap is in how strongly each organism responds, not in whether it could — both carry oxygen-responsive regulatory sites at iron genes.

Three Errors, One Cause

Categories were first built by keyword matching. It failed three times — each time the pattern matched text that was not gene function.

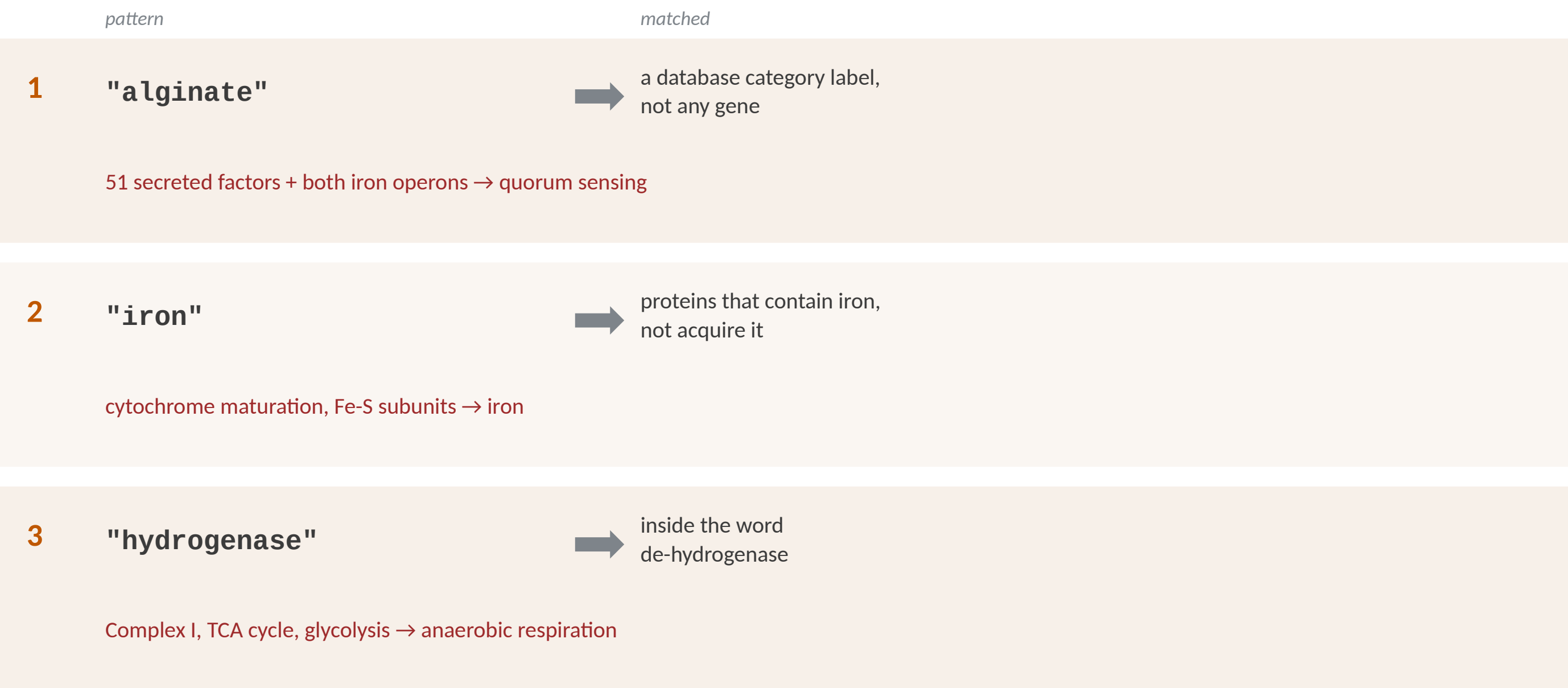


Figure 5. The third was found only by independent review, after the sets had already been audited twice. Separately, *arcA* names two unrelated genes in these organisms — matching sets by name would have created a false agreement.

What This Means

The conserved iron response is not new on its own — the iron-sparing program is well described in single organisms. What the matched design adds is the comparison. It shows the program holds across lineages while the oxygen response does not, and it does so with the perturbation held constant, which single-organism studies cannot do.

The practical implication is for anyone reading host-infection transcriptomes, where a bacterium is short of iron and oxygen simultaneously. These results say the iron half of that signature should look similar between species while the oxygen half will not — so an iron signature seen in one organism cannot be assumed to mean the same thing in another.

Limitations

- The *Pseudomonas* iron treatment (PQS) is both an iron chelator and a quorum-sensing signal, so that column is not a clean iron perturbation. A closely related signal, HHQ, produced no detectable response at the same dose — which argues against a purely quorum-driven explanation, but the confound stands and is disclosed rather than argued away.
- Both species carry oxygen-responsive regulation at iron promoters; *P. aeruginosa* has an ANR site alongside Fur and BqsR sites upstream of the ferrous transporter. The finding concerns response magnitude under these conditions, not regulatory capability.
- The *Shigella* datasets publish only genes reaching significance, so no non-significant background exists and enrichment testing was not possible for those cells.
- Two biological replicates per *Pseudomonas* condition; one contrast used a variance-shrinkage moderated t-test approximating limma rather than limma itself.
- The hypothesis was revised once after the first dataset was examined, so the *Pseudomonas* findings are exploratory. The category freeze preceded all *Shigella* analysis.
- Three keyword artifacts were found and corrected, all of one kind: a pattern matching text that was not gene function. A later completeness audit added 47 genes that met a category definition but had never been assigned. No revision changed a result.

Acknowledgments & References

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